AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

# Week 7: Digital Battery Passport

Mon 14 Dec - Fri 18 Dec 2026 · Sprint 7 · Overall progress 78%

| | | | |
|---|---|---|---|
| **Student name** | | **Roll no.** | |
| **Faculty in charge** | | **Date submitted** | Fri 18 Dec 2026 |

**On track** **This week in one line:** Built a tamper-evident passport: every event is SHA-256 chained and Ed25519 signed. 8 passports with 164 events verify; all 5 tested attacks are detected.

## 1. Goal for the week

Record each battery's whole life in a passport that anyone can verify and no one can secretly change, aligned with EU Regulation 2023/1542.

## 2. Day-by-day work

| Day | Work done |
|---|---|
| Mon 14 Dec | Schema: 10 event types, 6 actor roles (manufacturer, OEM, BMS, assessor, repurposer, recycler), required fields, public and restricted access. |
| Tue 15 Dec | Ed25519 key pairs per actor; canonical JSON hashing; signed, hash-chained ledger. |
| Wed 16 Dec | Verification of chain links, content hashes, signatures, role authorisation, required fields and chronology. |
| Thu 17 Dec | Public (QR-code) view hiding restricted health data; attack tests. Fixed a bug where signed data could be changed by the caller. |
| Fri 18 Dec | Issued passports for all 8 cells, added the live tamper demo to the website, this report. |

## 3. Completed deliverables

- **batterylife/passport/**: schema.py, crypto.py, ledger.py, store.py, issue.py.
- **results/passports/**: 8 full and 8 public passports.
- tests/test_passport.py (10 tests).

## 4. Results

| Attack on a passport | Detected by |
|---|---|
| Raise a past SOH value | Content hash |
| Raise a value and recompute every hash | Ed25519 signature |
| Delete an event | Chain link |
| Back-date an event | Chronology check |
| Re-sign with another actor's key | Signature |

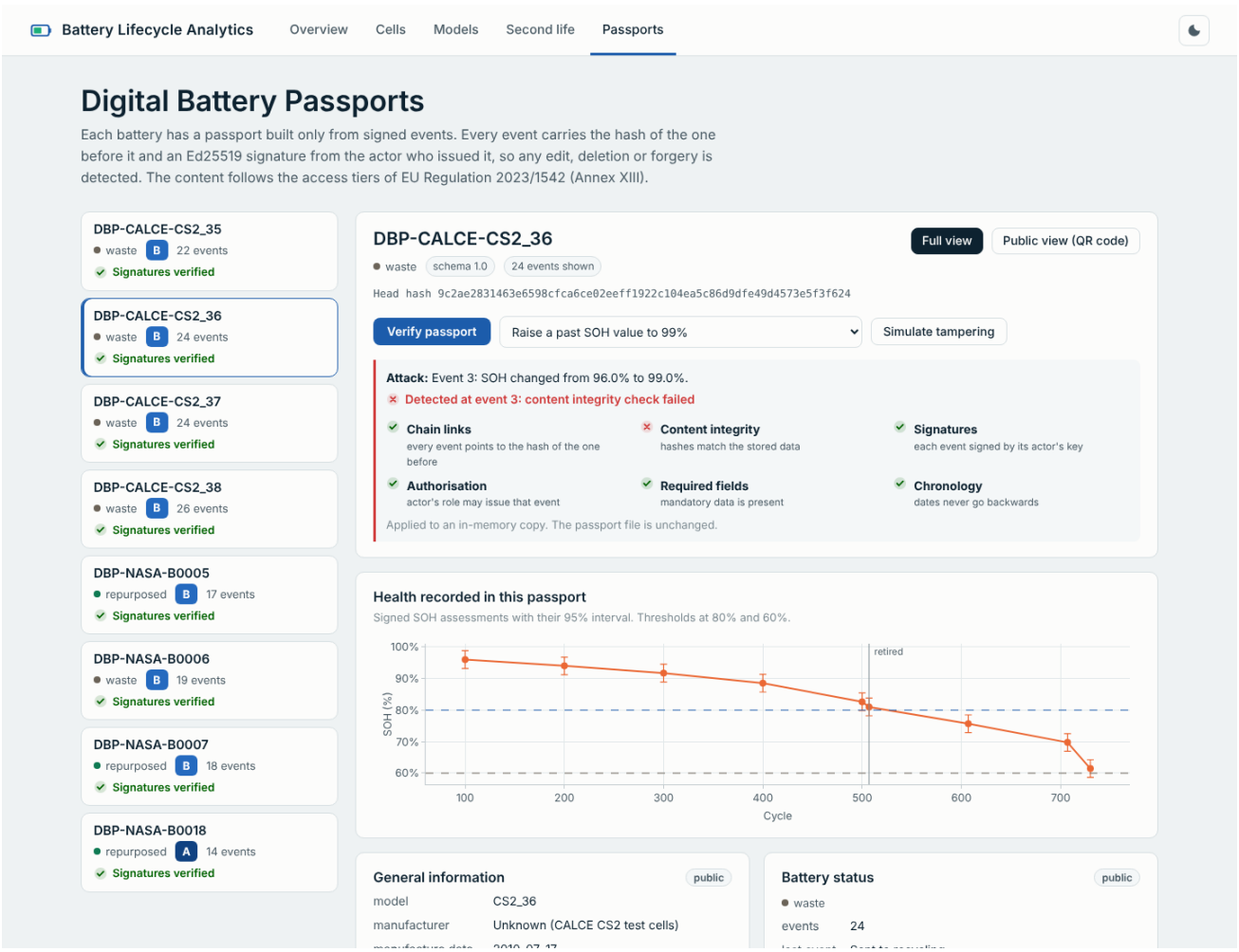



Figure 7. Website passport view: a forged SOH value is caught at event 3 and highlighted in the ledger.

## 5. Problems faced and how they were solved

| Problem | Resolution |
|---|---|
| Signed event data was shared with the caller's object. | Data is deep-copied at signing time; a test guards it. |
| Timestamps went backwards after repurposing. | Administrative events now follow by hours; chronology is a verification check. |
| A single-store ledger cannot stop deletion of a whole file. | Future work: anchor each chain head in a public registry. |

## 6. Plan for next week

- Integrate all stages into one pipeline and command-line tool.
- Pipeline tests and a full reproducibility run.

**Faculty remarks**

Faculty signature: _____ Student signature: _____